

PHY201: Waves & Optics

Problem Set 2

August 20, 2025

1. A spring balance consists of a pan that hangs from a spring. A damping force $F_d = -bv$ is applied to the balance so that when an object is placed in the pan, it comes to rest in the minimum time without overshoot. Determine the required value of b for an object of mass 2.5 kg that extends the spring by 6.0 cm. (Assume $g = 9.81 \text{ m/s}^2$.)
2. An undamped oscillator has a natural frequency $\omega_0 = \pi \text{ rad/s}$. Various amounts of damping are added to the system to give values of the damping factor γ equal to 0.01, 0.30 and 1.0 s^{-1} , respectively. (a) For each value of γ find the corresponding Q value and frequency ω of the damped oscillations. Comment on the change in ω over this range of γ . (b) For each of the values of Q , use a spreadsheet program to plot $x = A_0 \exp(-\gamma t/2) \cos \omega t$ over the time period $t = 0$ to $t = 10 \text{ s}$, using a value of 10 mm for A_0 . (c) Obtain an expression for x for the case of critical damping with the initial conditions, $x = 10 \text{ mm}$ and $dx/dt = 0$. Plot x over the time period $t = 0$ to 10 s .
3. According to classical electromagnetic theory, an accelerating electron radiates energy at a rate $P = Ke^2 a^2 / c^3$, where a is the acceleration, e is the electronic charge, c is the velocity of light and K is a constant with a value of $6 \times 10^9 \text{ N m}^2 \text{C}^{-2}$. Suppose that the motion of the electron can be represented by the expression $x = A \sin \omega t$ during one cycle of its motion.
 - (a) Show that the energy radiated during one cycle is $Ke^2 \pi \omega^3 A^2 / c^3$.
 - (b) Recalling that the total energy of a harmonic oscillator is $\frac{1}{2} m \omega^2 A^2$, where m is the electron mass, show that the quality factor is $Q = mc^3 / Ke^2 \omega$.
 - (c) Using a typical value of ω for a visible photon, estimate the lifetime of the radiating system. (Take $e = 1.6 \times 10^{-19} \text{ C}$, $m = 9.1 \times 10^{-31} \text{ kg}$.)
4. Many oscillatory systems, although their loss or dissipation mechanisms is not analogous to viscous damping, show an exponential decrease in their stored *average* energy with time $\bar{E} = \bar{E}_0 e^{-\gamma t}$. A quality factor Q for such oscillators may be defined using $Q = \omega_0 / \gamma$, where ω_0 is the natural angular frequency.
 - (a) When the note “middle C” on the piano is struck, its energy of oscillation decreases to one half of its initial value in about 1 s. The frequency of middle C is 256 Hz. What is the Q of the system?
 - (b) If the note an octave higher (512 Hz) takes about the same time for its energy to decay, what is its Q ?
 - (c) A free, damped harmonic oscillator, consisting of a mass $m = 0.1 \text{ kg}$ moving in a viscous liquid of damping coefficient b (with viscous force $F_{\text{viscous}} = -bu$), and attached to a spring of spring constant $k = 0.9 \text{ N/m}$, is observed as it performs damped oscillatory motion. Its average energy decays to $1/e$ of its initial value in 4 s. What is the Q of the oscillator? What is the value of b ?